Lesson 9 Teacher's Key: On the Scoreboard

001 Game Creator (Windows only) - Instructor Reference Only

VIDEO GAME CREATION

What This Lesson Teaches

001 Game Creator's actual, wiki-documented persistent-UI mechanism: an Interface containing Fields, distinct from the one-off Message Box used in Lessons 6 and 8. This was specifically researched and verified against the official wiki before writing this lesson, given this engine's history of needing extra care.

Level 1 - Full Worked Solution

- An Interface is a scene-level, persistent UI layout - fundamentally different from Message Box, which is a one-off popup. Interfaces are the correct, documented tool for HUDs in this engine (confirmed via the official 001 Game Creator wiki's Field page, and the wiki's own "RPG – Menus/HUD" tutorial, which uses this exact mechanism for a health display).
- Each Field's "Use Values" section can bind directly to a Variable via "Retrieve Text/Value" - and critically, ticking "Update Real-time" makes that binding live, refreshing automatically the instant the Variable changes, with no manual re-set action needed anywhere in the Trigger scripts.
- The score increment itself lives in the enemy's existing "Collided with actor" Trigger, alongside the enemies_remaining decrement - no new Trigger needed, since the display updates itself via the real-time binding.

Highlighted lines are new this week; everything else is code you already have.

```
enemy Actor Trigger: Collided with actor ->
  Variable – Modify Value(score, Add, 10)
  Variable – Modify Value(enemies_remaining, Subtract, 1)
  If(enemies_remaining <= 0)
    Message Box("You Win!")
  Actor – Destroy/Revive Actor(relate, Destroy)
  Actor – Destroy/Revive Actor(this, Destroy)

Interface Field (Score): Use Values -> Retrieve Text/Value -> Variable:
score, Update Real-time: ON
Interface Field (Lives): Use Values -> Retrieve Text/Value -> Variable:
lives, Update Real-time: ON
```

Why This Structure

- This is meaningfully different from every other engine's HUD approach this week: 001 Game Creator's binding is neither push (an explicit function call after each change) nor

exactly poll (no per-frame check written by the student) - the engine itself handles keeping the display in sync, once Update Real-time is configured correctly.

- This is exactly why the setup step matters so much this week: get the Field's binding and Update Real-time setting right once, and the Trigger-script side of this lesson becomes almost trivially simple (one new Variable action in an existing Trigger).

Level 2 / Level 3 Talking Points

- This is worth explicitly contrasting with GDevelop's Text objects this same lesson: GDevelop needs an explicit "always" event to push text updates every frame, while 001 Game Creator's real-time Field binding handles that automatically once configured - two different engines solving the same problem with genuinely different underlying mechanisms.
- If time allows, demonstrate what happens with Update Real-time UNCHECKED (the Field freezes until the Interface is closed and reopened) - seeing the broken behavior makes the correct setting's purpose click for students immediately.

Level 2 - complete code:

```
// Level 2 is verification only - the Level 1 code is already correct, so
no lines change.
// Checks: 3 kills read exactly "Score: 30"; an escape moves lives but
never score;
// both numbers are correct the instant the game starts, not only after
the first kill or escape.
```

Level 3 - complete code (all goals combined):

Highlighted lines are the Level 3 additions on top of the Level 1 code.

```
enemy Actor Trigger: Collided with actor -> (unchanged, still the only
place score changes)
  Variable – Modify Value(score, Add, 10)
  Variable – Modify Value(enemies_remaining, Subtract, 1)

Interface Field (Score) - Level 3: label text "Score: \\V[score]" with a
styled font (formatting markup instead of a raw bound value)
Interface Field (Lives) - Level 3: a Units Field type with a heart icon,
bound to lives
```

Common Student Mistakes

- Forgetting to tick Update Real-time on one or both Fields - the single most likely mistake this week, and one that looks like a mysterious bug rather than an obviously missing setting.
- Forgetting to set the Interface to display automatically at game/level start (a separate setting from the Field bindings themselves).
- Binding a Field to the wrong Variable (e.g., both Fields accidentally bound to score) - always test both independently, by shooting an enemy alone and separately letting one escape alone.